

Activity 14

Practicing code and creating visualizations in a Quarto file

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Armed Forces Data Wrangling Redux

Table 1: Army Enlisted Ranks by Gender

rank	female	male
Corporal OR Specialist	14619	81278
First Sergeant OR Master Sergeant	1426	9287
Private	4705	26672
Private First Class	8169	38802
Sergeant	11111	55671
Sergeant First Class	4322	30367
Sergeant Major OR Command Sergeant Major	413	2908
Staff Sergeant	7432	50030

Table 1 displays the distribution of enlisted Army personnel by rank and gender. Across each rank, the total males outnumber the females heavily. For example, Corporal or Specialists have over 81,000 males but only 14,619 females. This pattern is similar in other ranks, with the largest gaps appearing in mid-tier ranks. In terms of independence, it can be concluded that rank and sex are not independent of each other because the table highlights the decrease of female representation as the rank increases. There is a significant association between the two.

Popularity of Baby Names

Figure 1: A Time-Series Line Chart of Name Popularity Over Time

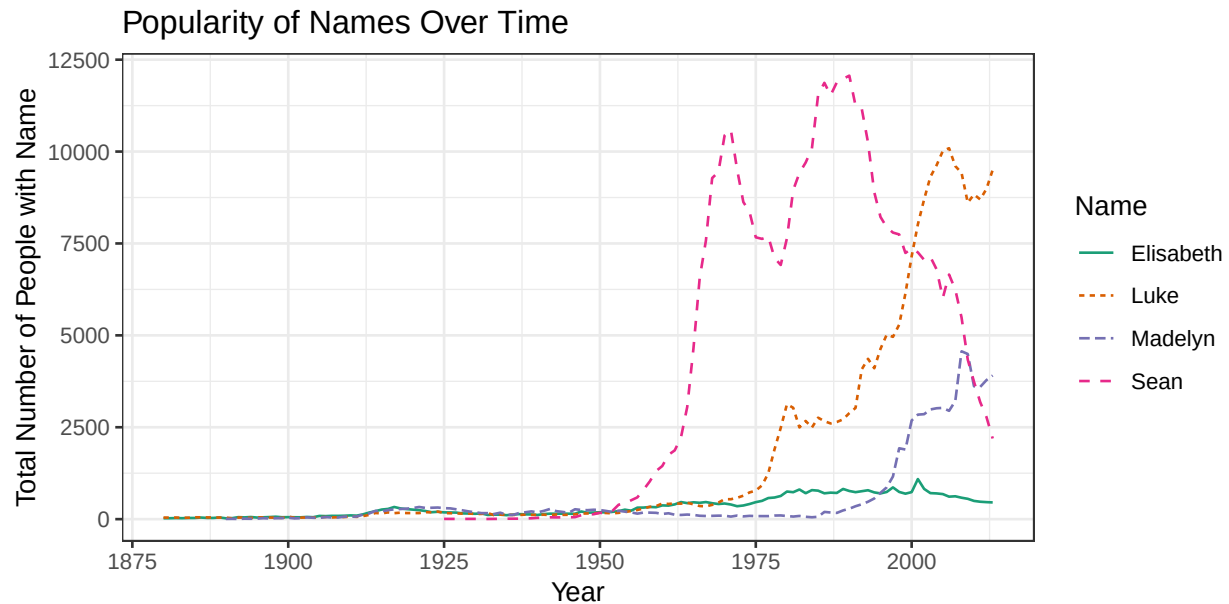


Figure 1 displays the count of four baby names over time: *Elisabeth*, *Madelyn*, *Sean*, *Luke*. Each name is displayed by a colored, patterned line. The visualization shows that Sean and Luke are the most popular amongst the 4. Each name started to increase after 1950. Sean spikes in the 1960's and 1990's but has decreased since then while Luke continues to rise. Madelyn spikes in 2010 while Elisabeth has remained in the lower count. These trends illustrate how name popularity changes with time and the trajectories they can follow. The names were chosen to include the author's name and her three other siblings.

Plotting a Mathematical Function

Figure 2: A Lineplot of Square Cut Size and the Volume of a Box it creates

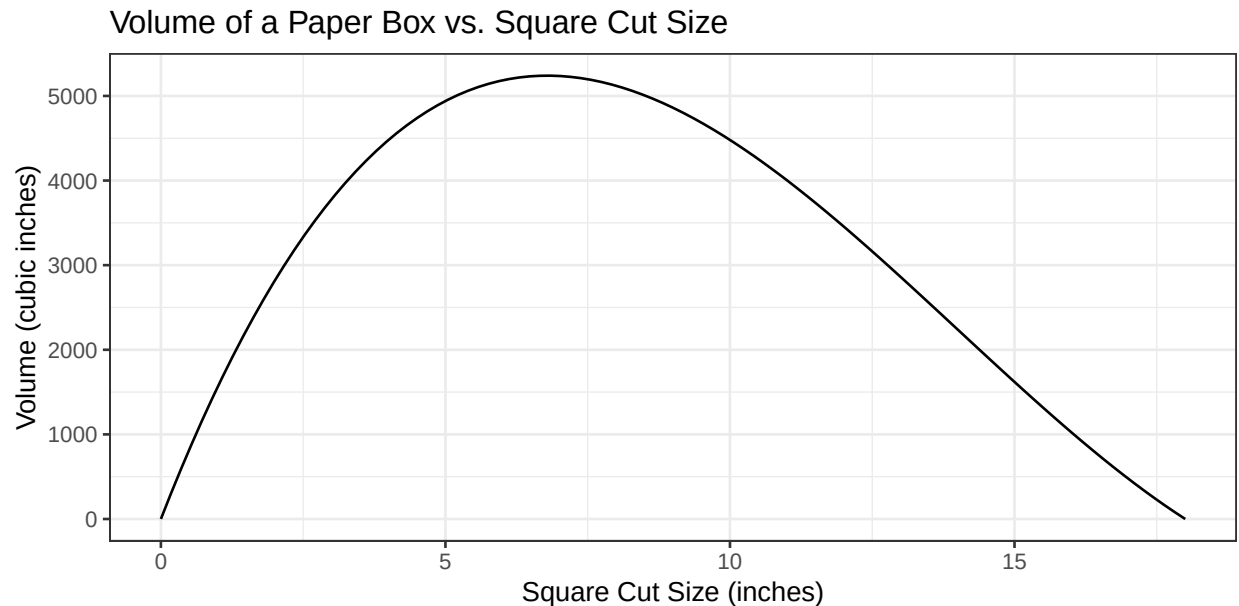


Figure 2 shows how the volume of a box created from a 36-in by 48-in piece of paper varies depending on the square cut size. The line starts at 0, rises to a peak and falls again to 0. The peak indicates the largest volume of the box. The visualization helps see the relationship between cut size and box volume. To answer the original box problem question: *what's the max volume and for what side length*, The chart peaks at around 6 inches which corresponds to a box volume of about 5,100 cubic inches.

What I Have Learned so Far

In this course so far, I have learned to plan my code before writing it by identifying the goal, needs, and necessary steps, making the coding process more efficient and accurate. All of our assignments have required us to write a plan before we write code. I have learned to write well-organized and neat code by adding comments, properly naming and assigning variables, and neatly structuring the syntax and design of the code. I have gained an understanding of what tidy data is. It contains cases and variables. A case refers to a specific, unique observation. (e.g. girls named Sofia in 2020) A variable refers to a specific attribute or feature of the case. This might be *count* of each case or *sex* which would indicate the case if male or female. Through code, we learned how to clean and organize data, as well as how to analyze it to extract meaningful insights. I have learned to extract data from other sources like web scraping. For example, to create Table 1 I had to scrape data from 2 different links, a google sheet for Armed Forces Pay Grades and a website for Ranks. I tidied it so that each case is an individual and then took a subset to create a frequency table from. When it comes to visualizing data, I know how to determine which types of visualizations

are appropriate for different data, create various types of visualizations, and accurately describe and interpret what each visualization conveys.

Code Appendix

```
# TABLE 1 : Armed Forces Frequency Table - Army Enlisted
## Wrangle the data
## 1. Load needed packages ----
library(googlesheets4)
library(rvest)
library(tidyverse)
library(readr)
library(stringr)
library(janitor)
library(kableExtra)
library(knitr)

## 2. Load Armed Forces data ----
gs4_deauth()
ArmedForces_Raw <- read_sheet(
  ss = "https://docs.google.com/spreadsheets/d/1cn4i0-ymB1ZytWXCwsJiq6fZ9PhGLUvbMBHlZqG4bwo/ed",
  skip = 2,
  col_names = TRUE,
  na = c("", "N/A*")
)

## 3. Read in Pay Grade and Ranks data ----
ranks_paygrade_raw <- read_html('https://neilhatfield.github.io/Stat184_PayGradeRanks.html') %>%
  html_elements(css = "table") %>% ## 3a. Select appropriate elements ----
  html_table() ## 3b. construct the tables ----

### 3c. Identify and Extract target tables ----
ranks_paygrade_raw <- ranks_paygrade_raw[[1]]

## 4. Clean the data ----

### 4a. fix column names, make unique ----
names(ranks_paygrade_raw) <- make_unique(names(ranks_paygrade_raw))
names(ranks_paygrade_raw)[names(ranks_paygrade_raw) == ""] <- "junk"

ranks_paygrade_clean_ind <- ranks_paygrade_raw %>%
  ### 4b. rename columns ----
  rename(
    "pay_grade" = `Pay Grade`,
    "army" = `Ranks by Branch of Service`,
```

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"navy" = `Ranks by Branch of Service.1`,
"marinecorps" = `Ranks by Branch of Service.2`,
"airforce" = `Ranks by Branch of Service.3`,
"spaceforce" = `Ranks by Branch of Service.4`,
"junk1" = `Ranks by Branch of Service.5`
) %>%
select(-matches("^junk\\d*$")) %>% ### 4c. remove unneeded columns ----
filter(pay_grade != "Pay Grade") %>% ### 4d. remove unneeded rows ----
filter(pay_grade != "Note: -- indicates that a pay grade is not currently used by a service br
pivot_longer(cols = 2:6, ## 4e. stack rank columns ----
             names_to = "branch",
             values_to = "rank")

## 5. Clean Armed Forces Data ----
armed_forces_clean_ind <- ArmedForces_Raw %>%
  ### 5a. Rename Columns ----
  rename(
    'pay_grade' = `Pay Grade`,
    'army_male' = Male...2,
    'army_female' = Female...3,
    'junk-0' = Total...4,
    'navy_male' = Male...5,
    'navy_female' = Female...6,
    'junk-1' = Total...7,
    'marinecorps_male' = Male...8,
    'marinecorps_female' = Female...9,
    'junk-2' = Total...10,
    'airforce_male' = Male...11,
    'airforce_female' = Female...12,
    'junk-3' = Total...13,
    'spaceforce_male' = Male...14,
    'spaceforce_female' = Female...15,
    'junk-4' = Total...16,
    'junk-5' = Male...17,
    'junk-6' = Female...18,
    'junk-7' = Total...19
  ) %>%
  select(-matches("^junk-\\d*$")) %>% ### 5b. remove columns not needed ----
  ### 5c. remove rows not needed ----
  filter(!pay_grade %in% c("Pay Grade", "", "Total", "Total Enlisted", "Total Warrant Officers",
    pivot_longer(cols = 2:11, ### 5d. stack branch_gender columns ----
                 names_to = "branch_gender",
                 values_to = "count") %>%
    separate_wider_delim(cols = branch_gender, ## 5e. separate column into branch and gender ----
                         delim = "_",
                         names = c("branch", "gender")) %>%
  mutate(

```

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    count = as.numeric(str_replace_all(count, ",", "")), ### 5f. convert count to numeric and
    count = replace_na(count, 0) ### 5g. replace NA with 0 ----
  ) %>%
# 6. Uncount the count column ----
uncount(weights = count)

## 7. Join the data ----
armed_forces_ind <- left_join(
  x = ranks_paygrade_clean_ind,
  y = armed_forces_clean_ind,
  by = join_by(pay_grade, branch)
)

## Make the armedforcesFrequency Table
armed_forces_ind %>%
  filter(branch == "army", str_detect(pay_grade, "^E")) %>% ### choose E subgroup
  janitor::tabyl(rank, gender) %>% ### tabulate sex and rank
  kable(
    booktabs = TRUE,
    format = "latex"
  )

# FIGURE 1 : Baby Names Popularity
## Make a lineplot of name popularity

### Load needed packages
library(dcData)
library(ggplot2)
library(tidyverse)

### Load the data
data("BabyNames")

### Create the filtered dataframe
BabyNames_4 <- BabyNames %>%
  filter(name %in% c("Elisabeth", "Madelyn", "Sean", "Luke")) %>% ### select 4 names
  group_by(year, name) %>% ### group by year and name
  summarise(total = sum(count)) ### sum the count column so it adds male and female into one

BabyNames_4 %>%
  ggplot( ### map the aesthetics
    mapping = aes(
      x = year,
      y = total,
      color = name,
      linetype = name
    )
  ) +

```

```

geom_line() + ### creates a line plot
scale_color_brewer(palette = "Dark2") + ###color blind friendly
theme_bw() +
labs( ### add labels
  x = "Year",
  y = "Total Number of People with Name",
  color = "Name",
  linetype = "Name",
  title = "Popularity of Names Over Time",
  alt = "A time series line chart showing the popularity trends of the names Elisabeth, Made
)
# FIGURE 2 : Plotting Mathematical Function
library(ggplot2)
## Create largest_volume function
largest_volume <- function(square_length, paper_length = 36, paper_width = 48){
  volume <- (paper_length - 2 * square_length) * (paper_width - 2 * square_length) * square_length
  return(volume)
}
## Plot the function
ggplot(data.frame(x = c(0, 18)), aes(x = x)) + ### Set the scale of the x axis
stat_function(fun = largest_volume) + ### Applies the function to the each axis value
  labs( ### add labels
    title = "Volume of a Paper Box vs. Square Cut Size",
    x = "Square Cut Size (inches)",
    y = "Volume (cubic inches)",
    alt = "Lineplot showing the volume of a box based on different square cut sizes. The square
  ) +
  theme_bw()

```